International Journal of Precision Engineering and Manufacturing-Green Technology (2025) 12:1363–1384
https://doi.org/10.1007/s40684-025-00730-3

Online ISSN 2198-0810
Print ISSN 2288-6206

REGULAR PAPER

Carbon Emissions Prediction and Optimization Method of Hobbing with Multi-source Data Collaborative Based on Federated Learning

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Received: 30 October 2024 / Revised: 25 February 2025 / Accepted: 1 March 2025 / Published online: 28 March 2025
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Abstract

Sustainability represents a foundational element of Industry 5.0, with low-carbon manufacturing representing a pivotal route to sustainable manufacturing. The Data-driven approach of process parameter optimization for low carbon control is an effective means to reducing carbon emissions during the manufacturing process. However, it relies on large amounts of valid historical data. To overcome the challenge of insufficient valid data within individual enterprises and the necessity for safeguarding data privacy during cross-enterprises collaboration, this paper proposed a method leveraging federated learning to enable multi-source data collaboration across enterprises for training a predictive model of hobbing carbon emissions. Addressing the issue of non- independent identically distribution (non-IID) characteristics inherent in mutil-source hobbing carbon emission data, and the inefficacy of traditional federated learning methods on such complex datasets, this paper designs an improved federated learning algorithm, Federated increment Attentive Message Passing (Fed-I-AMP), to construct the mapping model between hobbing process parameters and carbon emissions. Subsequently, a multi-objective optimization model was developed with the objective of minimizing carbon emissions and Comprehensive Evaluation Index (CEI). This model was solved using the multi-objective Coati algorithm (MOCOA). Experimental validation demonstrates that this approach significantly enhances the convergence performance and prediction accuracy of the model. Additionally, the optimized process parameters provide scientific evidence and practical guidance for low-carbon hobbing operations.

Keywords Gear hobbing · Carbon emission prediction · Federated learning · Parameter optimization

1 Introduction

China's total carbon emissions reached a staggering 12.6 gigatons in 2023 [1], making it the largest carbon emitter globally and facing great pressure to reduce emissions. The manufacturing sector serves as a pillar industry in China, contributing 33.6% of the country's total carbon emissions in 2021 [2]. Notably, energy consumption for mechanical processing alone accounts for over 70% of the total energy consumed by manufacturing [3]. Hobbing is the most widely used technology in gear manufacturing [4]. Hobbing machines is the largest retained in gear processing machine tools, characterized by low energy utilization efficiency, significant carbon emissions, and numerous influencing factors. Notably, process parameters significantly impact the carbon emission of hobbing machines [5], thereby presenting substantial potential for optimization to reduce emissions. Achieving low-carbon optimization in hobbing processes holds significant importance for reducing carbon emissions in the manufacturing sector and advancing the nation's dual-carbon goals [6].

Low-carbon methods at the manufacturing unit level include the three major categories of optimized process control [7], process/machine tool selection [8–11], and optimized machine tool design [11]. Among these, optimizing process parameters can be realized without altering the machine equipment during the manufacturing process [12], which is an important part of the low-carbon control of the manufacturing process. Traditional approaches primarily delve into the relationship between process parameters, machining energy efficiency, and carbon emissions through experimental design [13] and theoretical analysis [14]. With

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the promotion and application of machine learning in the manufacturing sector, data-driven low-carbon optimization methods have gradually become a research hotspot in academia [15]. Xiao et al. considered various factors such as data volume, time granularity, feature selection, and algorithm performance, and conducted a comparative analysis of multiple machine learning and deep learning methods for energy efficiency prediction [8]. Huang et al. constructed the mapping relationship between process parameters and machine energy consumption and lay-up factor by CNN-LSTM in carbon fiber reinforced polymer drilling processing [9]. He et al. proposed an adaptive energy consumption prediction method based on deep learning to address the problems of low efficiency and poor generalization of manually selected features [10].

Building accurate, reasonable, and reliable prediction models through data-driven approaches is a crucial prerequisite for optimizing process parameters, which is based on the collection of a large number of high-quality data in the production process [16]. However, most enterprises are lacking in digitalization and informatization [17], and hobbing is typically conducted in a multi-variety, small-batch production mode [18], which significantly increases the challenge of gathering sufficient production data, thereby constraining the construction of highly precise prediction model. Scholars address the issue of insufficient samples through strategies such as data augmentation, feature engineering, and algorithm optimization. Lu et al. established a hybrid dataset incorporating simulation and experimental data for milling processes and proposed an integrated transfer learning model for predicting cutting energy consumption [19]. Bai et al. analyzed feature impacts on model performance using Recursive Feature Elimination and Cross-Validation, then selected significant features to improve classification accuracy [20]. Wang et al. enhanced the generalization of tool wear prediction models under small sample conditions by optimizing the hyperparameters of domain generalization models through meta-learning [21]. Kumar et al. introduced additional sparsity costs into the cost function of convolutional neural networks and fine-tuned the model parameters through transfer learning [22]. Nenchev et al. optimize artificial neural networks and Gaussian process regression via XGboost, thereby enhancing the performance of the models on imbalanced classes and small, unevenly distributed datasets [23].

Against the backdrop of the thriving Industrial Internet and the increasingly refined division of labor among enterprises, the concept of cross-enterprise collaboration and its supporting technology system have demonstrated exploratory progress and practical applications in the manufacturing sector [24]. Currently, cross-enterprise collaboration primarily focuses on the coordination of manufacturing resources, production task scheduling [25]. In actual production environments, multiple enterprises may receive similar hobbing production tasks, and their machining centers will accumulate substantial hobbing production data with common characteristics. In view of this, for gear hobbing manufacturing enterprises with insufficient data at the initial stages of intelligentization, leveraging the Industrial Internet to achieve data sharing and collaborative modeling among multiple enterprises represents an effective pathway to overcome the limitation of insufficient data samples within a single enterprise. Production data, as an information resource of great strategic value, is an important cornerstone for facilitating enterprise informatization and making rational decision [17]. However, traditional approaches have relied on centralized methods to aggregate data from multiple manufacturing enterprises [26]. Furthermore, there are relatively few laws and patents related to protecting data [27]. Additionally, the mining production data has the potential to divulge confidential information regarding a company's business operations, production plans, manufacturing processes, and other pertinent details, posing a severe threat to corporate security. While data processing technologies can be employed to protect data privacy [28], the reality is that data is strictly prohibited from leaving a company's private servers [29], making the direct sharing of data for cross-enterprise collaborative modeling less feasible. Therefore, exploring novel methods that can both ensure data privacy and enable cross-enterprise collaborative modeling is a key issue in advancing the process of enterprise intelligence.

Federated learning, as an innovative distributed machine learning paradigm, implicitly transfers knowledge across multiple datasets through shared models, thereby achieving efficient collaborative modeling among multiple parties while protecting data privacy [30]. And it has practical applications in many engineering fields, including additive manufacturing and bearing fault diagnosis. Manan et al. have utilized federated learning to predict feature dimensions and evaluate the suitability of components in additive manufacturing [31]. Deng et al. proposed a federated learning framework integrated with domain adaptation, enhancing processing parameter planning for aircraft structural parts amidst data discrepancies among enterprises [32]. Ning et al. have designed federated support vector machines and federated random forest algorithms tailored for horizontal and vertical federated learning, respectively, constructing predictive models for production line failures [33].

Previous work has demonstrated that data-driven approaches can effectively mining the relationship between process parameters and machining carbon emissions. However, the accuracy of model predictions highly contingent upon the quantity and quality of training data, which presents a critical obstacle to the application of machine learning in enterprises during the early stages of intelligence adoption. Federated learning can address the issue

of insufficient data in individual enterprises by enabling cross-enterprise collaborative model training while protecting corporate data privacy. Hobbing carbon emission data is often complex and diverse, with poor inter-correlation in practical industrial scenarios. To date, no research has been conducted on the realization of cross-enterprise collaborative prediction of hobbing carbon emissions in the context of non-IID hobbing datasets. Thus, the main contributions of this paper are as follows:

(1) Based on the requirements of cross-enterprise collaboration and data protection, a method for leveraging federated learning to achieve multi-source data collaboration for constructing a hobbing carbon emissions prediction model is proposed.
(2) In response to the non-IID characteristics of multi-source hobbing carbon emissions datasets, the Fed-I-AMP algorithm is introduced to enhance the predictive performance of the model.
(3) A multi-objective optimization model targeting comprehensive efficiency and carbon emissions is proposed to balance the dual demands of low-carbon manufacturing processes and corporate profitability. The optimal parameters for low-carbon hobbing operations are obtained through the solution derived from the improved MOCOA algorithm.

The subsequent content of this paper is organized as follows: Sect. 2 constructs a carbon emission model for hobbing processes and analyzes the characteristics of multi-source hobbing carbon emission datasets. Section 3 describes the improvement strategies for federated learning methods under the condition of non-IID datasets. Section 4 presents the multi-objective optimization model and the corresponding optimization algorithm. Section 5 conducts experimental verification and comparative analysis. Finally, Sect. 6 provides conclusions and perspectives.

2 Hobbing Carbon Emissions Modeling and Analysis of Mutil-Source Datasets

The carbon emission characteristics of hobbing processes across different enterprises are similar, but the influencing mechanisms are complex, resulting in the coexistence of both similarities and differences within multi-source datasets. In Sect. 2.1, a staged model for collaborative carbon emission modeling and optimization in gear hobbing is introduced. Subsequently, in Sect. 2.2, the characteristics present among multi-source datasets are analyzed.

2.1 Staged Modeling of Carbon Emissions in Gear Hobbing

The carbon emission sources in gear hobbing process are numerous, and the emission mechanisms are complex, influenced by various factors such as working conditions, process parameters, and gear specifications. Additionally, the variations in production conditions among enterprises further complicate the calculation of carbon emissions during the manufacturing process. Based on the carbon emission boundary model * MERGEFORMAT [34], the carbon emissions from gear hobbing processes primarily consist of three components: energy consumption carbon emissions C_e, material consumption carbon emissions C_m, and waste disposal carbon emissions C_w.

Carbon emissions associated with waste disposal, such as gear hobbing scraps, occur after the completion of hobbing process and are not influenced by process parameters. Therefore, this component of carbon emissions is neglected in this study. Carbon emissions related to material consumption, such as cutting fluids, are more influenced by the manner of usage rather than process parameters * MERGEFORMAT [35]. Additionally, there exist significant differences among various enterprises in this regard. To reduce model complexity and enhance prediction accuracy, this paper adopts a staged modeling approach for hobbing carbon emissions. To reduce model complexity and enhance prediction accuracy, this paper adopts a staged modeling approach for gear hobbing carbon emissions. Specifically, during the cross-enterprise collaborative prediction stage, only C_e, which is strongly correlated with process parameters, is considered. However, in the process parameter optimization stage, both C_e and C_w are considered.

(1) C_e refer to the carbon emissions from the electricity consumed during the hobbing process, which is one of the main sources of carbon emissions during the production of gears. The calculation formula for C_e is as follows:

$$C_e = E_e \times EF_{elec} \tag{1}$$

where E_e is the total electricity consumption of hobbing process, and EF_{elec} is the electric emission factor. It is important to note that due to significant differences in emission factors across different regions; to mitigate the impact of this factor, a fixed value should be adopted during carbon emission accounting. The formula for calculating the total energy consumption of hobbing is as follows:

$$E_e = E_{st} + E_{air} + E_{cut} \tag{2}$$

standby stage E_{st}:The carbon emissions in the standby stage are mainly derived from the electric energy consumed by the CNC hobbing machine during standby, which is affected by factors such as the performance of the machine tool itself

and the proficiency of workers' operation. The calculation formula can be expressed as:

$$E_{st} = \int_0^{t_{st}} P_{st} d_t \tag{3}$$

where t_{st} is the standby time and P_{st} is the standby power.

Air cutting stage and tool retiring stage E_{air}: The characteristics of this stage are high instantaneous power but short duration, accounting for a small proportion of the total carbon emissions of hobbing. The calculation formula can be expressed as:

$$E_{air} = \int_0^{t_{air}} (P_u + P_{auc} + P_{st}) d_t \tag{4}$$

where P_u represents the no-load power, which has a quadratic function relationship with the technological parameters such as the spindle speed and feed rate of the machine tool * MERGEFORMAT [5]; P_{auc} represents the processing associated auxiliary system power, determined by the performance of the machine tool itself.

cutting stage E_{cut}: the power of the cutting stage is calculated as:

$$E_{cut} = \int_0^{t_{cut}} (P_u + P_{auc} + P_{st} + P_a + P_c) d_t \tag{5}$$

where P_a is the additional load power, which is quadratically related to the cutting power P_c.

The total carbon emission of hobbing machining in the optimization stage consists of two parts: electrical energy carbon emission and material consumption, and its formula can be expressed as:

$$C_{total} = C_e + C_t = C_e + C_{tool} + C_{pcf} \tag{6}$$

where C_{tool} is the carbon emission due to tool wear and C_{pcf} is the carbon emission due to cutting fluid consumption.

(2) C_{tool} are defined as the carbon emissions generated during the preparation of the hobbing cutter, averaged over the completion of each hobbing process * MERGEFORMAT [36]. The calculation formula is as follows:

$$C_{tool} = \frac{t_{cut}}{T_{tool} \times (R+1)} \times m_{tool} \times EF_{tool} \tag{7}$$

where m_{tool} is the mass of the hob, EF_{tool} is the carbon emission factor of the hob, t_{cut} and T_{tool} represent the total time of a single hobbing process and the working time of the hob before tool change, respectively, and R represents the grinding times of the hob. As derived from the above formula, C_{tool} are determined by the hob itself and its preparation process on the one hand and are also indirectly affected by process parameters on the other hand.

C_{pcf} are defined as the averaged over the carbon emissions produced over a single hobbing operation from the preparation and disposal of cutting fluid

$$C_{pcf} = \frac{t_{cut}}{T_{tool}} \times (m_o + m_\alpha) \times EF_{pcf} \tag{8}$$

where T_{pcf} is the replacement cycle of the cutting fluid, M_o is the initial volume of the cutting fluid, M_α is the additional volume of the cutting fluid during its life cycle, and EF_{pcf} is the carbon emission factor of the cutting fluid.

2.2 2Analysis of Multi-source Carbon Consumption Datasets

2.2.1 Similarity Analysis

Hobbing is a multi-blade continuous cutting process for gear manufacturing. As shown in Fig. 1, the cutting process of hobbing can generally be divided into four stages: standby (A), air cutting (A-C), cutting (C-G), and tool retraction (G-A). Although different enterprises employ various machine tools for hobbing, the hobbing process itself is relatively fixed, with corresponding process parameters highly overlapping. The key process parameters during this process include spindle speed, feed rate, and machining allowance. Consequently, the hobbing datasets from different enterprises show a high degree of similarity in structural [5, 37].

The real-time power curve during the hobbing process is shown in Fig. 2. The blue A and red B curves describe the actual instantaneous power observed during the hobbing operations of various hobbing machines across different enterprises. The black curve represents a theoretical fitting curve of the hobbing power, segmented according to the various processing stages. Although there are certain differences in the instantaneous power and total energy consumption values of hobbing processes under different models and process parameters, the influence patterns of process parameters on the machining power remain similar. From the perspective of the cutting phase, as the depth of

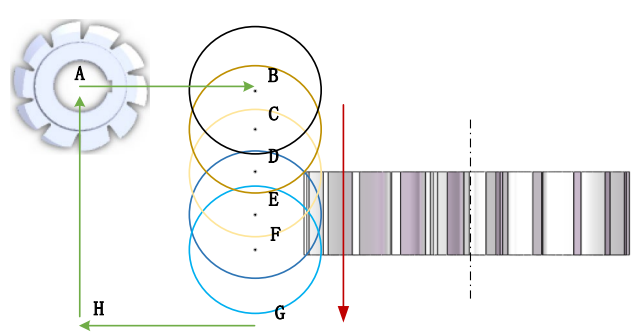

Fig. 1 Schematic diagram of hobbing process

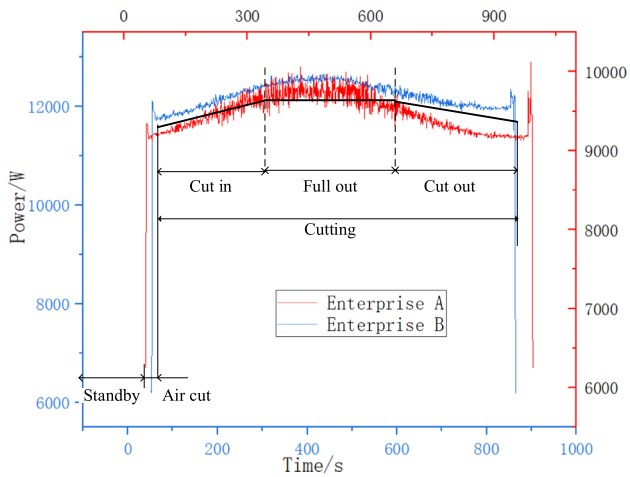

Fig. 2 Real-time power curve of gear hobbing

cut increases, the actual power curves of various hobbing machines exhibit a linear increase with the increment of the depth of cut. Upon attainment of the maximum depth of cut, maintain a relatively stable state. Subsequently, as the depth of cut decreases, the power begins to gradually decline. Consequently, the datasets from different enterprises also exhibit similarity in content.

Based on the above analysis, the hobbing process is fixed, coupled with the similarity in data types such as process parameters and gear specifications, results in a high degree of overlap in the feature dimensions of hobbing datasets. Furthermore, the power variation patterns across different hobbing processes also exhibit similarities. This structural and content similarity within hobbing datasets can support collaborative modeling research utilizing multi-source data.

2.2.2 Difference Analysis

Differences between multi-source hobbing carbon emissions datasets are inevitable. Corporate production is order-oriented, the order directly determines the output of a particular specification gear, manifesting as differences in dataset size among various enterprises, which can be expressed as $n_1 \neq n_2 \neq \cdots \neq n_{client}$. Simultaneously, enterprises determine the appropriate process to meet the requirements of product specifications and performance. Small-modulus gears typically require a single pass of hob roll cut [18], whereas large-modulus and high-precision gears often necessitate two or more passes [38]. This disparity in production processes results in differences in the characteristics of the collected datasets, which can be expressed as $x_1 \neq x_2 \neq \cdots \neq x_{client}$. Furthermore, the carbon emissions associated with hobbing processes constitute complex, multi-factorial, and dynamically characteristics. Taking the emissions of the energy consumption component as an example, the energy consumption characteristics vary significantly among machines of different specifications. Additionally, even within the same machine, discrepancies in energy consumption data may also be observed due to variations in maintenance status and operating conditions, such as tool wear, which can be expressed as $P_1 \neq P_2 \neq \cdots \neq P_{client}$ [39].

The impact of disparities among carbon emissions datasets from multiple enterprises can be categorized into the following points: 1) Inconsistency in the amount of data due to orders or demand. The party with fewer data points may be recognized as noise by intelligent algorithms during the training process, thereby reducing the model's generalization ability [30]. 2) Inconsistency in the data dimensions due to differences in the process. Differences in the dimensions of data arising from various production workflows can affect the weighting of features during training. These disparities in feature dimensions may even preclude feature alignment, rendering the training process unfeasible [40]. 3) Inconsistent data distribution due to the differences in hobbing machine beds, gear types, etc. Variations in data distributions can lead the model to learn in multiple directions simultaneously, thereby complicating the convergence process [41].

In summary, the multi-source hobbing carbon emissions datasets exhibit typical non-IID characteristics. However, it cannot simply be assumed that the differing datasets are unusable. The disparities in data can lead the model to learn in diverse directions. Benign weights can gradually steer the model towards convergence in the optimal parameter space, whereas malicious weights may deviate the model from it. The data from a single enterprise often fails to encompass a comprehensive range of gear types, and the range of process parameters set for the machining of similar gear types is limited. Conversely, multi-source data possesses strong complementary qualities [42], where diverse datasets can fill in the gaps in the coverage of gear types and process parameters within a single enterprise's dataset. So as to enhances the prediction accuracy and generalization performance of the model. Overcoming the numerous implications arising from the non-IID nature of mutil-source hobbing datasets is a major challenge to achieve cross-enterprise collaborative prediction of hobbing carbon emissions.

3 Carbon Emissions Prediction Method Based on Federated Learning

To achieve the collaborative prediction of carbon emissions in hobbing processes utilizing multi-source data, Sect. 3.1 introduces the collaborative training framework based on federated learning. In Sect. 3.2, considering the non-IID

characteristic of multi-source gear hobbing datasets, the original federated learning approach is improved.

3.1 Federated Learning-Based Framework

In the collaborative prediction of hobbing carbon emissions utilizing multi-source data, multi-source data refers to the data collecting from diverse enterprises. The client represents the actual hobbing production enterprises or manufacturers, who possess limited hobbing production data but sufficient computing power. Their goal is to train a prediction model for hobbing carbon emissions by leveraging the data from other members via the Industrial Internet of Things, to guide the actual hobbing processing of process optimization of process parameters. Hence, the problem can be formulated as follows: There are N_{client} clients, where the i-th client possesses a private dataset $D_i = \{(X,y)_j\}_{j=1}^{n_i}, i = 1,2,\cdots,N_{client}$. where n_i represents the number of samples in the i-th dataset, and $d_j = (X,y)_j$ denotes the j-th data point within this dataset, consisting of input features X and the predicted carbon emission value y.

The collaborative prediction of hobbing carbon emissions utilizing multi-source data is implemented based on federated learning, which has three primary types: horizontal FL, vertical FL, and federated transfer learning [43]. Among them, the FedAVG algorithm [30] is the classic method to horizontal FL. Its core idea involves training models across multiple edge devices, where users transmit sanitized gradients or models to a server without leaking their raw data. The server aggregates the information uploaded by each client and then consolidates the model parameters from these edge devices into a central server by averaging them. This process updates the global model parameters on the central server, with the model aggregation computation formula given by:

$$f_g(w) = \sum_{k=1}^{K} \frac{n_k}{n} f_k(w) \tag{9}$$

where $f_k(w)$ is the k-th local model trained on a local dataset, while $f_g(w)$ is the global model after completing one iteration of updates. $n=\sum_{k=1}^{K} n_k$, stands for the total number of samples across all datasets participating in the training process. The collaborative training framework for multi-source data based on FL is shown in Fig. 3 below:

3.2 Algorithm Improvement under Non-IID Datasets

The traditional FedAVG algorithm aims to train a globally optimal prediction model. However, there are inherent differences in production environments and conditions among various enterprises. The FedAVG algorithm cannot meet the demand for individualization requirements of each client model under the non-IID multi-source datasets. Moreover, the inclusion of less relevant models may result in a reduction in the prediction accuracy and convergence capability of the global model. Therefore, this paper proposes the Fed-I-AMP algorithm that combines the EWC algorithm with the FedAMP algorithm.

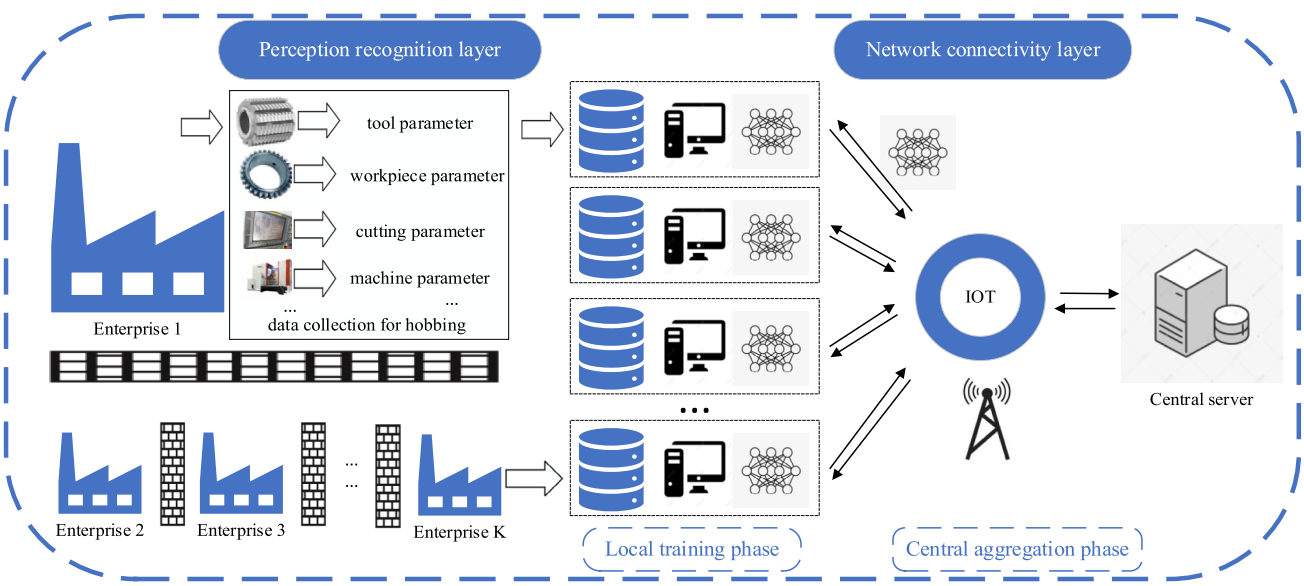

Fig. 3 Collaborative training framework

3.2.1 Parameter-consistent Approach to Continuous Learning

To mitigate the parameter differences encountered during model training on non-IID datasets, thereby enhancing model convergence and prediction accuracy, this paper introduces the Elastic Weight Consolidation (EWC) algorithm [44] during the local training phase to update the parameters of the prediction model. The key idea of EWC is to find a common parameter space for both old and new tasks during the training process [45], enabling the model further learning new knowledge from local datasets while retain knowledge from the original model as much as possible. In deep neural networks, the importance of weights can be calculated through first-order derivatives, with the calculation formula approximate as:

$$\omega^k = \left\| \frac{\partial L(f(X_k, w_g), y_k)}{\partial w_g} \right\| \tag{10}$$

where L is the loss function; for the local carbon emissions dataset D, the importance of the model weights ω^k is calculated using the samples (X_k, y_k) through the above equation. Subsequently, the importance matrix Ω of the parameters in the prediction model is derived.

$$\Omega_i = \frac{1}{|D|} \sum_{d \in D} w^k = \frac{1}{|D|} \sum_{d \in D} \frac{\partial^2 L(d, w)}{\partial w^2} \tag{11}$$

The loss function of the neural network in the local training phase is calculated as:

$$L_{all} = L_{local}(w_k) + L_{old}$$

$$L_{all}(w) = L_{local}(w_k) + \lambda_1 \sum_{i,j} \Omega_{i,j} \left(w_{l,i,j} - w_{g,i,j} \right)^2 \tag{12}$$

where λ_1 is the weight adjustment factor for the global model trained locally as a parameter change, a user-set hyperparameter, $L_{local}(w_k)$ is the loss of the carbon depletion prediction model on the local dataset, and w_l and w_g are the weight parameters of the local model and the global model, respectively. With the above Eq. 12, the model can be forced to learn the distribution of the local model $f(w_l)$ while preserving the knowledge of the global model $f(w_g)$, thus mitigating the divergence of the weights of the local and global models, thus ensuring better generalization performance and prediction accuracy.

3.2.2 personalized Approach to Federal Aggregation

Federated attentive message passing (FedAMP) represents a personalized federated aggregation strategy. Its key idea is leveraging the attention mechanism to evaluate the similarity between prediction models from different clients and the target client's model, subsequently assigning varying weights accordingly. This method enables the personalized updating of the target model [46]. The optimization objective function is formulated as:

$$min \left\{ G(W) := \sum_{i=1}^{K} F_i(w_i) + \lambda_2 \sum_{i<j}^{K} A\left(\left\| w_i - w_j \right\|^2 \right) \right\} \tag{13}$$

where $W = [w_1, \cdots, w_m]$ is the parameter set of the prediction model and the parameters of each client's prediction model, respectively. λ_2 is the Regularization weight. The total loss of FedAMP comprises two parts: the sum of the training losses $f(w)$ from each client's model, and the attention-induced function $A(w)$ that calculates the effectiveness of collaboration between clients. To ensure the convergence performance of the model, $A(w)$ is generally defined as a negative exponential function, as shown in Eq. 14.

$$A\left(\left\| w_i - w_j \right\|^2 \right) = 1 - e^{\|w_i - w_j\|^2 / \sigma} \tag{14}$$

where $\|w_i - w_j\|^2$ represents the Euclidean distance between two prediction models. The parameter σ is a user-defined hyperparameter that controls the magnitude of $A(w)$.

The overall optimization objective of FedAMP can be decomposed into two alternating optimization steps. Firstly, $A(w)$ is computed through gradient descent:

$$U^k = U^{k-1} - \alpha^k \nabla A(W^{k-1}) \tag{15}$$

where α_k is the learning rate for gradient descent. U^k is the constructed proximal term, and the subsequent optimization step is performed using the proximal point algorithm.

$$W^k = \arg min F(W) + \frac{\lambda}{2\alpha_k} \left\| W - U^k \right\|^2 \tag{16}$$

where $U^k = \left[u_1^k, u_2^k, \cdots u_m^k \right]$. Instead of simply averaging the model parameters based on the number of samples in the dataset, the model aggregation process in this scenario performs a weighted update according to the similarity between the models' parameters. Therefore, the update formula for the parameters of each client model can be rewritten as:

$$\xi_{i,j} = \alpha_k A' \left(\left\| w_i^{k-1} - w_j^{k-1} \right\|^2 \right) \tag{17}$$

$$u_i^k = \left(1 - \alpha_k \sum_{i \neq j}^{m} \partial A'\left(\left\|w_i^{k-1} - w_j^{k-1}\right\|^2\right)\right) \cdot w_i^{k-1} + \alpha_k \sum_{i \neq j}^{m} \partial A'\left(\left\|w_i^{k-1} - w_j^{k-1}\right\|^2\right) \cdot w_j^{k-1} = \xi_{i,1} \cdot w_i^{k-1} + \cdots + \xi_{i,m} \cdot w_m^{k-1} \tag{18}$$

where $\xi_{i,j}$ is the combination weight between the i-th and j-th prediction models, and the linear combination weights for the same model satisfy $\xi_{i,1} + \cdots + \xi_{i,m} = 1$. According to the properties of $A(w)$, it can be ascertained that $A'(w)$ is positive and decreases as w increases. The smaller Euclidean distance between similar prediction models results in higher weights being assigned to models similar to the target client during aggregation, thereby promoting collaboration among enterprises with similar datasets.

Combining the contents of 3.2.1 and 3.2.2, the FedIAMP algorithm, which is a method to realize the prediction of carbon emissions from hobbing gears with the synergy of data from multiple sources, is as follows.

Algorithm 1 Fed-I-AMP

mapping model between gear hobbing process parameters and electricity consumption-related carbon emissions.

(2) Comprehensive Evaluation Index: Solely focusing on reducing processing carbon emissions does not fully encompass the core concerns of enterprises. Therefore, it is necessary to conduct a collaborative optimization that considers processing cost M processing precision Q and production efficiency E. Therefore, this study achieves collaborative optimization by optimizing CEI [37], whose mathematical expression is given as follows:

$$CEI_i = w_1 \cdot \frac{M_i}{\overline{M}} + w_2 \cdot \frac{Q_i}{\overline{Q}} + w_3 \cdot \frac{E_i}{\overline{E}} \tag{19}$$

where the weighting coefficients w_1, w_2 and w_3 represent the respective weights of cost, accuracy, and efficiency in the comprehensive optimization objective.

For the processing cost M, this study focuses specifically on the components related to process parameters. The processing cost comprises three parts: direct cost (determined by electricity consumption, denoted as M_{dir}), indirect cost

4 Multi-objective Optimization Method Based on MOCOA

The main work of this chapter is to reduce hobbing process carbon emissions through optimize process parameters. In Sect. 4.1, the optimization objectives, optimization variables, and constraint conditions are defined. Subsequently, in Sect. 4.2, a multi-objective coati optimization algorithm (MOCOA) is employed to solve the formulated multi-objective optimization mathematical model.

4.1 Multi-objective Optimization Model for Hobbing Carbon Emissions

4.1.1 Optimization Objectives

(1) carbon emissions: The hobbing carbon emissions is calculated using the above formula 6, which is the primary optimization objective in this study. Specifically, the portion of carbon emissions related to electricity consumption is addressed through the utilization of federated learning, which enables cross-enterprise data collaboration to construct an associative

(stemming from material consumption, denoted as M_{indir}), and additional cost introduced by carbon tax policies (denoted as M_{add}). The calculation formula is as follows:

$$M = M_{dir} + M_{indir} + M_{add}$$

$$M = k_1 \cdot E_e + k_2 \cdot \frac{t_c}{T_{hob}} \cdot C_{hob} + k_3 \cdot C_{fluid} + k_4 \cdot C_{total} \quad (20)$$

where k_1 to k_4 are cost coefficients, t_c represents the processing time, and T_{hob} is the total life of the cutting tool. C_{hob} denotes the cost of the hob, and C_{fluid} represents the cost of the cutting fluid.

Radial runout F_r is one of the critical precision parameters for gear design and manufacturing. This study uses this indicator to characterize the machining precision Q of gears.

To ensure consistency in the numerical trends of all optimization indicators, this study represents total processing time as processing efficiency E. After such processing, a lower value of CEI indicates a better alignment with the enterprise's multifaceted requirements in terms of both economics and efficiency.

To eliminate the influence of different orders of magnitude, it is necessary to standardize the above optimization objectives. In terms of weight assignment, enterprises can determine the weights based on their specific preferences and strategic orientations. This study employs the entropy weight method, an objective weighting strategy, to calculate the entropy value of each indicator to assess the amount of information it carries, thereby determining the weights. This approach ensures that the optimization scheme not only aligns with the enterprise's actual needs but also possesses high operability and scientific rigor.

4.1.2 Optimization Variable

Through collaborative training, the mapping relationship can be established between process parameters for various workpieces, machining equipment and the hobbing carbon emissions. However, in the hobbing process of a real enterprise, the hobbing machines and the type of gears being processed are relatively fixed. Therefore, for a target enterprise focusing on a specific type of gear hobbing, the controllable variables mainly include the spindle speeds and feed rates for rough hobbing and finish hobbing, denoted as $[n_1, f_1, n_2, f_2]$, which are the four cutting parameters that can be adjusted.

4.2 Constraint Conditions

In actual hobbing processes, the specifications and performance of equipment such as machine tools and hobs impose constraints on the setting of cutting parameters. To ensure the stability of the hobbing process and the precision of the machined workpieces, the process parameters must satisfy the following constraint conditions:

(1) Spindle speed constraint:$n_{min} \le n \le n_{max}$. where n_{min} and n_{max} are the minimum and maximum allowable spindle speeds, respectively, for the gear hobbing machine.

(2) Spindle feed constraint:$f_{min} \le f \le f_{max}$. Where f_{min} and f_{max} are the minimum and maximum allowable spindle feed rates, respectively, for the gear hobbing machine.

(3) Maximum cutting force constraint: $F_c \le F_c^{max}$. where F_c^{max} is the maximum allowable cutting force for the gear hobbing machine.

(4) Machine power constraint:$p_{cut} \le \eta p_e$. where p_{cut} is the actual power consumption during machining, p_e is the rated power of the gear hobbing machine, and η is the cutting power coefficient.

(5) Surface roughness constraint:$0.0321 f_z^2 / r \le [Ra]$. Where r is the nose radius of hob, $[Ra]$ is the acceptable maximum surface roughness of the finished product.

(6) Tooth profile error constraint:$F_z^2 sin\alpha / 4d \le f_{cx}$. Where f_{cx} is the required tooth curve error.

4.3 Modified Multi-objective Cotia Optimization Algorithm

Coati Optimization Algorithm is a metaheuristic algorithm inspired by the natural behavior of coatis preying on iguanas [47]. Its optimization performance has been shown to surpass traditional metaheuristics across various benchmark functions. MOCOA builds upon the original algorithm by incorporating an archiving mechanism to store non-dominated individuals encountered during the optimization iterations [48]. It further employs a hypercube grid method to select elite individuals that guide the update process. The detailed flow of MOCOA is described as follows.

(1) Population initialization.

The size and distribution of initial population directly influence the convergence speed and solution accuracy of algorithms in solving multi-objective optimization models. The original COA algorithm generates the initial population randomly through a random number seed, which can easily lead to an uneven distribution of the initial population and trap the model in local extrema. To address this issue, this paper introduces the Good Point Set method [49] to initialize the algorithm's population, ensuring a more uniform distribution of the population within the search space, thereby reducing the risk of the algorithm getting stuck in local optimal solutions. The formula for this initialization method is as follows:

$$x_{i,j} = (ub_j - lb_j) \cdot \mathrm{mod}(r_s^{(N)} \times k) + lb_j \quad (21)$$

where $x_{i,j}$ represents the j-th dimension of the i-th coati individual, ub_j and lb_j are the upper and lower bounds,

respectively, for that feature under the given constraints, and n is the population size. r_s is the good point and its corresponding calculation formula is as follows:

$$r_s = \{2\cos(2\pi k/p), 1 \le k \le s\} \tag{22}$$

(2) coatis' strategy when attacking iguanas.

In the context of COA design, the coati exploration phase is subdivided into two distinct stages: eviction and search. Half of the coati population engages in evicting iguanas from trees, which can be represented mathematically as Eq. 23:

$$\begin{aligned} &x_{i,j}^{new} = x_{i,j} + r \cdot (I_{Igu} - I \cdot x_{i,j}) \\ &for\, i = 1, 2 \sim \lfloor N/2 \rfloor,\ and\, j = 1, 2, \sim, m \end{aligned} \tag{23}$$

The remaining half of the coati population preys on the iguanas that have fallen to the ground, which can be mathematically described using Eq. 25:

$$Igu_j^G = lb_j + r \cdot (ub_j - lb_j), j = 1, 2, \sim, m \tag{24}$$

$$\begin{aligned} &x_{i,j}^{new} = \begin{cases} x_{i,j} + r \cdot \left(Igu_j^G - I \cdot x_{i,j}\right), F < F_i \\ x_{i,j} + r \cdot \left(x_{i,j} - Igu_j^G\right),\ else \end{cases} \\ &for\, i = \lfloor N/2 \rfloor + 1, \sim, N,\ and\, j = 1, 2, \sim, m \end{aligned} \tag{25}$$

where r is a random number within the interval [0,1], t denotes the iteration number, and I is a random integer drawn from the set {1,2}. F is the fitness function, Igu_j^G represents the position of the iguana and the position of the optimal individual of the current coati population, corresponds to the optimal set of process parameters. This two-stage predation strategy, consisting of eviction and search, enables the coatis to continuously move towards promising solutions within the search space, reflecting the global search capability of the COA in the problem domain.

(3) coatis' escape strategy from predators.

When coatis are both hunting for prey and simultaneously need to evade predators, this behavior can be conceptually represented or modeled by formula 27:

$$lb_j^{local} = lb_j/t, ub_j^{local} = ub_j/t, \text{where } t = 1, 2, \sim, T \tag{26}$$

$$\begin{aligned} &x_{i,j}^{new} = x_{i,j} + (1 - 2r) \cdot \left(lb_j^{local} + r \cdot \left(ub_j^{local} - lb_j^{local}\right)\right) \\ &for\, i = 1, 2, \sim, N,\ and\, j = 1, 2, \sim, m \end{aligned} \tag{27}$$

where lb_j^{local} and ub_j^{local} represent the local lower and upper bounds of the decision variables, respectively, which are gradually decreasing as the iteration round t increases. This natural behavior of the coatis, where they move around the solution set, demonstrates the local search capability of the COA algorithm.

The two phases above both lead to changes in the coatis' position. If the updated individual is superior, it will replace the current individual; otherwise, the current individual remains unchanged, preserving the best-known solution thus far.

$$x_{i,j} = \begin{cases} x_{i,j}^{new}, F_i^{new} < F_i \\ x_{i,j},\ \text{else} \end{cases} \tag{28}$$

where X_i^{new} is the updated position.

5 Case Study

This section validates the proposed method for collaboratively constructing a carbon emissions prediction model for hobbing processes utilizing multi-source data and the subsequent optimization of a multi-objective model. Section 5.1 outlines the process of collecting the original hobbing carbon emissions data and presents the results of data pre-processing. In Sect. 5.2, the effectiveness of training the prediction model with multi-source data collaboration is verified. Section 5.3 delves into the investigation of factors influencing the performance of the prediction model trained with multi-source data collaboration. In Sect. 5.4, the optimal hobbing process parameters are obtained by solving the multi-objective optimization model through MOCOA.

5.1 Data Acquisition and Pre-processing

To validate the effectiveness of the proposed method in this paper, validation experiments were conducted in actual hobbing production workshops. Production data were collected from two enterprises: a machine tool factory and a transmission workshop, both located in Chongqing. These workshops encompass three machine tools, specifically, YS3120CNC6, YS3132CNC6, and YS3140CNC6. The models and parameters of these machine tools are outlined in Table 1 below. The process of data acquisition is shown in Fig. 4. Firstly, information about gear type and associated parameters is retrieved and recorded from the factory database. Subsequently, current sensors and alligator clips are installed within the power distribution box of the hobbing machine. The HIOKI PW3390 power analyzer is then employed to acquire real-time power measurements during the hobbing process. Concurrently, the corresponding cutting parameters are recorded based on the numerical control code utilized during the machining operation.

The data ultimately collected encompasses a range of gear design parameters such as modulus (m), number of teeth (z), tooth width (b), pressure angle (α), helix angle (β), addendum

coefficient (*ha**), clearance coefficient (c*), and modification coefficient (x). Additionally, the equipment parameters are mainly hobbing machine type. The cutting parameters primarily consist of the spindle speeds (n_1, n_2) and feed rates (f_1, f_2) for rough and finish hobbing, and the machining allowance (d). The remaining data includes energy consumption and processing time during gear hobbing, which are primarily used for calculating the carbon emissions of the hobbing process.

The process parameters for carbon emissions in hobbing possess distinct dimensions and units, which can significantly impact the results of data analysis. To eliminate the dimensional effects between different features and ensure consistent analysis, it is necessary to standardize the hobbing carbon emissions data. The Min–Max normalization method is adopted, which scales the same feature metrics across multiple enterprise datasets to a uniform level, and the formula is as follows:

$$y = \frac{x - x_{\min}}{x_{\max} - x_{\min}} \tag{29}$$

where x_{max} and x_{min} represent the maximum and minimum values of the features in the dataset respectively, x represents the original value of the data and y represents the normalised value.

In addition to continuous features such as hob speed and feed rate, the hobbing carbon emissions dataset also contains discrete features like machine types, which cannot be directly

Table 1 Continuous optimization result

Parameter	Unit	YS3120CNC6	YS3132CNC6	YS3140CNC6
Maximum machining diameter	mm	200	320	400
Maximum machining modulus	mm	6	8	12
Maximum hob Angle	°	±45	±45	±45
Maximum hob axial movement	mm	100	100	200
Maximum speed of hob spindle	rpm	1200	1200	700
Axial feed velocity (Z-axis)	mm/min	2000	2000	3000
Radial feed velocity (X-axis)	mm/min	2000	2000	3000
Main motor power	kw	5.5	7.5	18

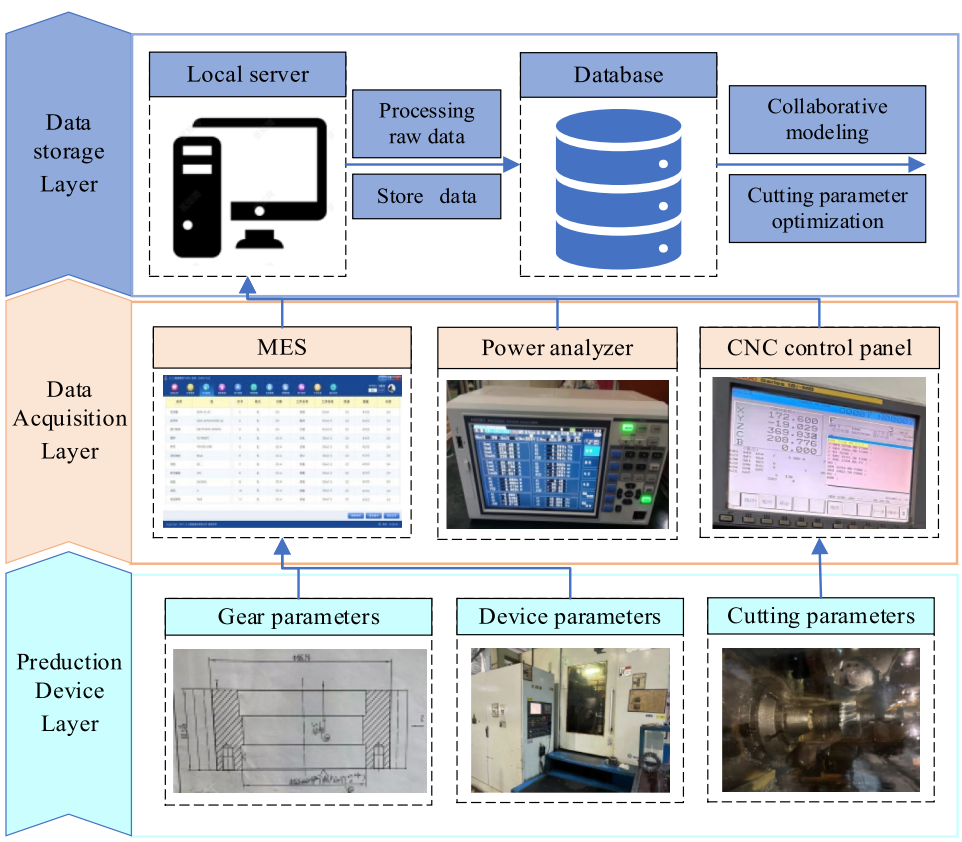

Fig. 4 Data Acquisition Flowchart

input into neural network models. To address this issue, One-Hot Encoding is employed to transform these categorical variables into a format that can be effectively processed by neural network models. For instance, two distinct hobbing machines can be represented by '01' for operations performed on one machine and '10' for the other.

Due to the varying feature dimensions among the hobbing carbon emissions datasets collected from different enterprises, it is necessary to align the features of the multi-source datasets involved in the collaborative training process. For a dataset $D = \{x_1, \sim, x_n, y\}$ which consists of n features, this paper utilizes Random Forest algorithm (RF) for feature selection. RF learns the corresponding decision rules from the hobbing carbon emissions dataset and evaluates the importance of each feature for carbon emissions using metrics such as information gain. Specifically, the weight of feature importance reflects the degree to which a feature contributes to the prediction results. By retaining significant features and eliminating those with lesser relevance, this approach aids in mitigating the risk of overfitting within the model. Moreover, the feature dimensionality reduction of multi-source hobbing carbon emission data set is beneficial to achieve feature space overlap, resulting in $D_1 = D_2 = \{x_1, \sim, x_m, y\} \subseteq D$. The feature importance ranking based on the decision tree is shown in Fig. 5 below.

It can be seen from the figure above, the primary features influencing the hobbing carbon emissions include the spindle feed rates during rough and finish hobbing, the spindle speeds during rough and finish hobbing, and the tooth width. Notably, excluding tooth width, the carbon emissions during the hobbing process are predominantly influenced by the process parameters, rather than the design parameters of the gears. Specifically, the feed rate of spindle and tooth width demonstrate a strong correlation with the machining duration, thereby exerting a more notable influence on carbon emissions. Spindle speed is more closely tied to machining power, whereas the servo motor power comprises only a minor fraction of the total power consumption in hobbing machines, thereby exerting a relatively lesser influence. Collectively, these five features contribute significantly to the carbon emissions in gear hobbing, with a cumulative contribution of more than 97%, effectively reflecting the information from the original high-dimensional feature set. Consequently, this paper employs these five features along with the machine tool type as input features to construct prediction model for hobbing carbon emissions with multi-source data collaboratively training.

5.2 Comparison of Different Methods and Results Analysis

The hobbing carbon emissions dataset, subjected to feature dimensionality reduction, is shown in Table 2, and parameters associated with the calculation of CEI are presented in Table 3 below. In total, more than three hundred pieces of data were collected and after removing duplicates the dataset has 86 pieces of data. Seventy of the data were used as the training set and partitioned into five subsets for training local models, The data volume of each training subset is 20, 15, 10, 15, 10 respectively, and they are obtained through random sampling without replacement from processed dataset. The remaining 16 are allocated as the validation set to verify the models. It is crucial to note that the approach considered in this study is tailored towards a single target user, thus there is no need to consider the scenario of adapting the target model to make it suitable for multiple target users.

The algorithm was implemented using PyTorch, a framework based on the Python. Backpropagation neural network is used as the base learner for local model training, optimized with the Adam optimizer. The values of the hyperparameters of the algorithm are shown in Table 4 below.

In this context, the "Network structure" refer to the number of layers in the neural network model, as well as the number of neuron nodes within each layer. Additionally, within the framework of federated learning, the term "local iteration round" in model training signifies the number of iterations a client model undergoes during one round of training on its local dataset. Meanwhile, "Global iteration round" denotes the number of communication rounds required to upload locally trained models to the cloud and aggregate knowledge across these models.

To validate the effectiveness of the hobbing carbon emissions prediction model, several evaluation metrics including Root Mean Square Error (RMSE), Mean Absolute Percentage Error (MAPE), Mean Absolute Error (MAE), and the Coefficient of Determination (R^2) were utilized to assess the predictive performance of the model. The respective calculation formulas for these metrics are provided as follows.

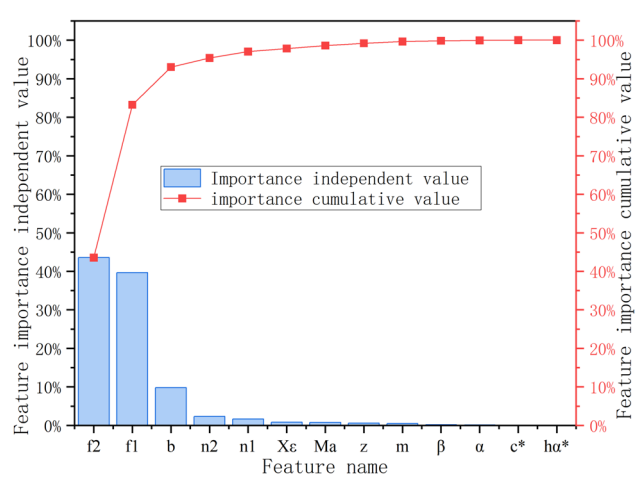

Fig. 5 Feature importance ranking

Table 2 Continuousoptimization result

No	Type	$n_1(rpm)$	$f_1(mm/min)$	$n_2(rpm)$	$f_2(mm/min)$	···	M(yuan)	CE(kgCO_2)
1	A	280	3.5	390	6	···	6.590895	2.482712
2	A	360	3.5	390	6	···	6.718729	2.574157
3	A	280	4	360	6	···	6.17543	2.349518
...	...	...	...	...	...	...	...	...
78	B	100	2.3	140	10	···	9.209392	3.921179
79	B	150	6	180	12	···	3.57418	1.326199
86	C	190	9.5	240	12	···	3.55654	1.312153

Table 3 Relevant parameters of CEI

Iteams	Values	Unit
EF_{elec}	0.5568	kgCO_2/(kW.h)
EF_{tool}	1.7	kgCO_2/kg
EF_{pcf}	3.05	kgCO_2/L
k_1	0.74	yuan/(kW.h)
k_2	800	yuan/kg
k_3	0.05	yuan
k_4	0.15	yuan/kgCO_2
w_1	0.47	/
w_2	0.11	/
w_3	0.42	/

Table 4 Hyperparameters set for Fed-I-AMP

	Iteams		Value
BPNN	Learning rate	Lr	0.01
	Activation function	/	'Sigmod'
	Network structure	/	8–20-20–15-1
Fed-I-AMP	Num of clients	N	5
	Global iteration round	/	50
	Local iteration round	/	100
	Weight regulation factor	λ_1	1000
	Regularization weight	λ_2	1
	Hyperparameter σ	σ	1

$$MAE = \frac{1}{n}\sum_{i=1}^{n}\left|y_i - \hat{y}_i\right|^2 \tag{30}$$

$$MAPE = \frac{1}{n}\sum_{i=1}^{n}\frac{(y_i - \hat{y}_i)^2}{y_i} \tag{31}$$

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(y_i - \hat{y}_i\right)^2} \tag{32}$$

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(\hat{y}_i - y_i)^2}{\sum_{i=1}^{n}(\overline{y_i} - y_i)^2} \tag{33}$$

where n is the total number of samples in the dataset used for testing, y_i is the true value of the i-th sample, and $\hat{y}_i$ is the predicted value of the i-th sample. Among the above evaluation metrics, a smaller value for RMSE, MAPE, and MSE indicates that the model can accurately establish the mapping relationship between input features and predicted values. $R^2 \in (-\infty, 1]$, with a coefficient closer to 1 signifies a better fitting performance of the model.

To validate the effectiveness of co-training in improving model predictive performance, this study sets up controlled experimental, and the framework contains the following training scenarios:

(1) *Solo Training Approach*: In this scenario, the target client trains the model solely using its limited data. This setting aims to simulate the realistic scenario of a single enterprise under data deficiencies.

(2) *Central Training Approach*: As an extreme counterpoint, this scenario disregards data privacy concerns and directly utilizes data from all enterprises to train the model. This scenario aims to simulate the upper bound of model performance under ideal conditions.

(3) *FL Training Approaches*: To evaluate the impact of co-training on model performance improvement and validate the effectiveness of the proposed Fed-I-AMP method, it is comparatively analyzed against three classical federated learning algorithms: FedAvg, FedProx, and FedAMP.

Figure 6 and Fig. 7 respectively show the iterative changes in loss and accuracy on the test set for several FL approaches during global knowledge aggregation. Due to the randomness in model initialization, there exist substantial differences in the initial loss values. It is observable that the loss functions of these FL methods gradually converge as the iteration process progresses, and Fed-I-AMP achieves the highest accuracy after convergence. Notably, the loss iteration curve for Fed-I-AMP differs significantly from other methods, which is attributed to the constrained learning

approach employed by Fed-I-AMP through EWC, resulting in a slower decline in loss values during the initial stages.

For convenience in subsequent text, the model trained by a particular method is referred to directly by the name of that method. In the series of illustrations shown in Fig. 8, Fig. 8(a) and Fig. 8(c) show the test set data alongside the carbon emission prediction values generated by various predictive models, while Fig. 8(b) and Fig. 8(d) show the corresponding prediction errors of these carbon emission models. It is evident that there exists no single model that achieves the lowest error across all data points, signifying the inexistence of an absolutely optimal and universally applicable model. However, the performance of a model can be evaluated holistically. As observed in Fig. 8(b), Solo exhibits the poorest predictive performance, manifested not only by significant prediction errors at individual data points but also by the largest number of data points with significant errors. In contrast, Central shows outstanding predictive performance, characterized by the largest number of data points with errors less than 0.025, and most points with prediction errors below 0.1. By comparing the prediction errors of various FL methods, it can observe that these methods generally exhibit lower error levels on the test set compared to the Solo model. This not only validates the effectiveness of the strategy of collaborative training of models using multi-source data, which enhances data diversity and scale for model training, but also confirms that such method can improve the predictive accuracy of carbon emissions models.

Table 5 presents the mean and variance of various indicators for the above methods on the test set across 10 experiments, aiming to evaluate and compare the predictive accuracy and stability of performance among these methods. It is evident that Solo exhibits significant fluctuations across all indicators, indicating that training model solely on a small

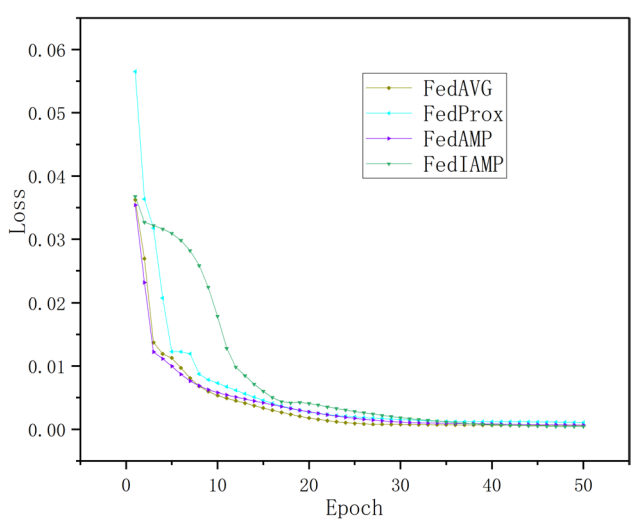

Fig. 6 Iterative loss graph

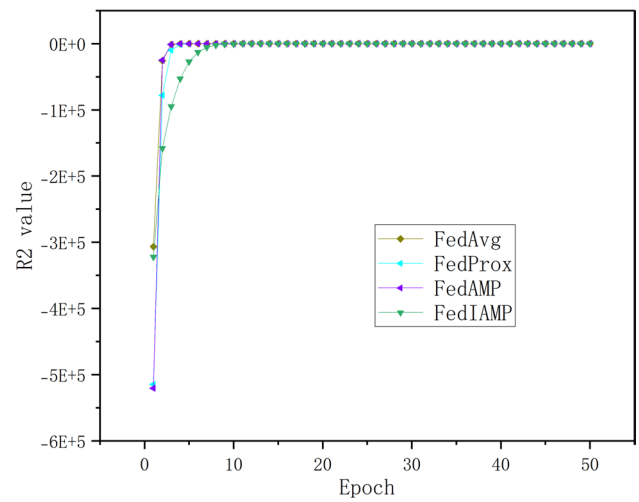

Fig. 7 Iterative precision graph

dataset fails to achieve stable and effective convergence, and is prone to overfitting issues. In contrast, all the indicators of the four FL approaches are significantly superior to Solo, highlighting the efficacy of the FL mechanism in aggregating knowledge across multi-source datasets to enhance the generalization ability and prediction accuracy of the models. Further comparison of the performance metrics on the test set among various FL approaches, it is evident that the proposed method, Fed-I-AMP, achieves an R2 score of 0.9498, which is slightly higher than the original FedAMP (0.9429), and notably superior to FedAvg (0.9306) and FedProx (0.9334), thereby enhancing the prediction accuracy of the model. Additionally, the standard deviation of the R2 score for Fed-I-AMP is merely 0.51, representing the lowest among the four methods, indicating a higher degree of consistency and stability in its performance.

However, it is imperative to note that despite the pronounced advantages of training models with multi-source data over single datasets, the model performance still lags behind the prediction accuracy and convergence achieved by the Central approach. This suggests that the FL framework, when integrating dispersed knowledge from multiple datasets, encounters performance losses and efficiency bottlenecks.

5.3 Analysis of Factors Affecting Model Prediction Performance

To fully validate the effectiveness and predictive performance of the improvement strategy of Fed-I-AMP, this section delves into and analyzes the following crucial factors.

(1) Distribution of data, type of hobbing contained in a single dataset: FL employs a distributed learning mechanism to enable multiple models to learn from their respective local

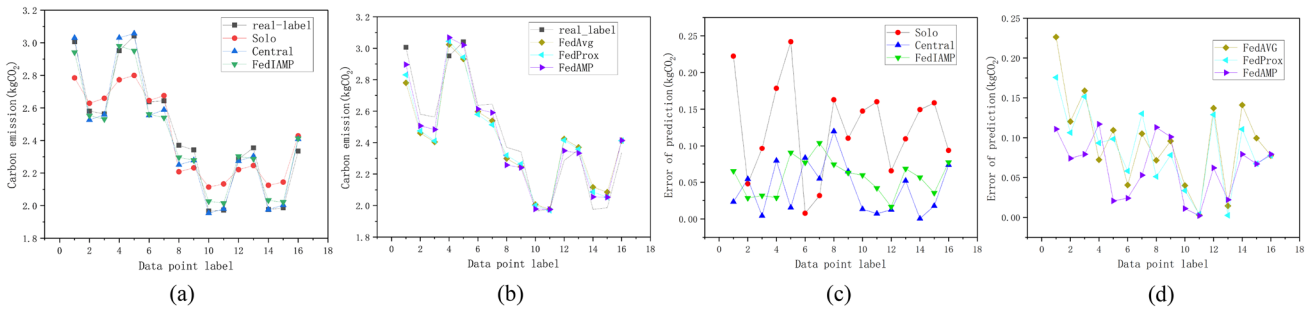

Fig. 8 Forecast Comparison Chart

Table 5 Performance Comparison Table

	MAE*10^{-2}	RMSE*10^{-2}	MAPE*10^{-2}	R2*10^{-2}
Solo	12.13 ± 6.13	14.66 ± 7.78	5.98 ± 2.98	69.99 ± 29.49
Central	4.03 ± 0.21	4.86 ± 1.19	1.88 ± 0.10	96.97 ± 0.55
FedAvg	6.74 ± 1.17	7.78 ± 1.42	3.20 ± 0.56	93.06 ± 2.50
FedProx	6.53 ± 0.95	7.68 ± 0.88	3.08 ± 0.46	93.34 ± 1.53
FedAMP	6.02 ± 0.87	7.12 ± 0.73	2.84 ± 0.43	94.29 ± 1.18
FedIAMP	5.21 ± 0.51	6.66 ± 0.88	2.46 ± 0.23	94.98 ± 1.27

datasets, storing and transmitting knowledge in the form of model parameters. In practical scenarios, particularly when collaborating across enterprises, the heterogeneity in machine tools and gear specifications adopted by different enterprises results in heterogeneity in data, which in turn leads to heterogeneity in model parameters. This poses challenges in converging parameters of multiple models into the optimal model when integrating their knowledge, significantly degrading the generalization performance of the models.

To evaluate the impact of the above differences in data distribution on model prediction accuracy, this study conducted a partitioning of the original dataset based on machine tool types and process parameters. In this study, dataset 1 originates from a YS3140CNC6, datasets 2 and 3 are sourced from YS3132CNC6, and datasets 4 and 5 are derived from YS3120CNC6. The size of each dataset is the same as those set in Sect. 5.2. Among them, Dataset 5 represents the target enterprise, while the other datasets serve as assisting parties. Except for Dataset 4, the datasets of the remaining assisting parties exhibit low similarity to the target enterprise dataset. Experimental validations were conducted on four FL methods, with the hyperparameters held constant, and the experimental result is presented in Table 6.

It can be observed that when multi-source datasets exhibit such extreme imbalance in distribution, the predictive performance of the three traditional FL methods decreases significantly. Specifically, FedAvg and FedProx, exhibit the most pronounced decline, with the MAE, RMSE, and MAPE metrics increasing nearly four times. Additionally, R^2 drops from 0.93 to -0.26 and -0.13, respectively, which not only indicates the loss of precise prediction capability by the models, but also reflects the inapplicability of these two traditional FL methods in such scenarios. In contrast, while FedAMP experiences a relatively smaller decrease in performance, Fed-I-AMP maintains a high level of predictive accuracy, demonstrating the strongest robustness compared to the other three algorithms.

Through analysis, the performance degradation of FedAvg and FedProx is determined by the mechanism of the models themselves. Specifically, datasets 1, 2, and 3 are more different from the target dataset, so there is greater heterogeneity in the parameters learned by each model in the local training phase. As shown in Eq. 9, both FedAvg and FedProx update the global model by using the sample size as the weighting coefficient. This strategy is prone to making the model susceptible to the influence of irrelevant models when applied to datasets with significant differences, thereby deviating from the optimal parameters space and leading to a notable decline in the prediction performance of the model. In contrast, FedAMP assigns weights by assessing the similarity between models, thereby enabling personalized updates for each local model. During this process, models with higher degrees of similarity are assigned larger weights, effectively facilitating model collaboration and knowledge fusion, while models with significant differences are assigned lower weights to effectively mitigate the risk of the target model deviating from optimal parameters space. However, the heterogeneity arising from the significant differences in model parameters due to the training of local models on diverse datasets still exists, ultimately leading to a decline in the predictive performance of the FedAMP model. Furthermore, Fed-I-AMP introduces EWC algorithm based on personalized model updates, which consolidates the target model's critical parameters for the target data while allowing the model to continue learning from different local datasets. This design enhances the consistency of important parameters between the target model and

Table 6 Prediction results of different federated learning models

	MAE	RMSE	MAPE	R2*10^{-2}
FedAvg	0.28 ± 0.11	0.32 ± 0.12	0.14 ± 0.06	− 0.26 ± 0.85
FedProx	0.26 ± 0.10	0.30 ± 0.10	0.13 ± 0.05	− 0.13 ± 0.64
FedAMP	0.14 ± 0.02	0.17 ± 0.02	0.07 ± 0.01	0.67 ± 0.08
Fed-I-AMP	0.05 ± 0.00	0.06 ± 0.01	0.03 ± 0.00	0.96 ± 0.01

collaborating models, enabling it to maintain high prediction accuracy even in non-IID datasets.

In summary, the distribution of local samples in the hobbing carbon emissions dataset is a crucial factor influencing the performance of federated learning models. In scenarios where the dataset exhibits extreme distribution and poor data correlation, the predictive performance of Fed-I-AMP significantly superior to the other three traditional federated learning models. Fed-I-AMP shows lower values across various error metrics, with MAE, RMSE, and MAPE being 0.05, 0.06, and 0.03 respectively. Furthermore, it achieves a superior fitting effect, evidenced by an R^2 value as high as 0.96.

(2) Number of data, number of multi-source datasets involved in co-training: FL enhances model predictive performance and generalization by integrating knowledge across multiple datasets. Within this process, the number of collaboratively trained carbon emissions datasets emerges as a critical factor influencing model prediction accuracy. To delve into the mechanisms of this factor, based on the dataset division in Sect. 5.2, we evaluate the predictive performance of various FL models under different data scales by adjusting the number of datasets participating in collaborative training, while keeping the test set and the target enterprise's training set constant.

Figure 9(a-h) shows the dynamic variations of performance metrics for four federated learning models across multiple experiments under varying dataset sizes. Specifically, Fig. 9 (a, c, e, g) present the mean changes in four key indicators: MAE, MAPE, RMSE, and R^2, for the carbon emissions prediction models on the test dataset. These metrics collectively reflect the predictive performance of the models. In contrast, Fig. 9 (b, d, f, h) show the variance changes in the above four performance metrics, which provide insights into the stability of the models' predictive performance. As evident from the above figure, with the datasets number increase, the prediction errors of the models constructed by the four FL methods all exhibit a decreasing trend, and the convergence capabilities of the models are also significantly enhanced. Notably, the most pronounced reduction in prediction error occurs when the data scale increases from two to three. Specifically, MAE for FedAvg and FedProx are 0.2298 and 0.2220 for two datasets, which decrease to 0.1222 and 0.1426 for three datasets, representing a decrease of 46.82% and 35.77%, respectively. However, when comparing these two models with Solo, it is observed that the prediction errors of both models increase from 0.1213 to 0.1222 and 0.1426, respectively.

Upon analysis, the significant increase in the predictive performance of the FedAvg and FedProx models in Scenario 2 can be attributed to the limitations of traditional FL methods when training on non-IID data. On one hand, the difference among datasets leads to heterogeneity in model parameters, and the models struggle to effectively complement and converge collaboratively during the training process in smaller datasets scenarios; On the other hand, the aggregation mechanism that assigns weights based on sample quantity fails to adequately account for differences between models, ultimately amplifying the prediction errors. However, as the dataset number expands and the knowledge base becomes richer, the models learn from diversified samples, thereby mitigating the challenges posed by data heterogeneity. Consequently, the models regain their ability to exhibit higher prediction accuracy.

In contrast, methods such as FedAMP and Fed-I-AMP, which consider model similarity as a weighting factor, can effectively mitigate convergence issues arising from excessive differences in model parameters, show notable superiority in handling non-IID data. Taking Fed-I-AMP method as an example, it outperforms the other three traditional FL methods across various error metrics in all four scenarios. Notably, even with smaller dataset numbers, the.

prediction errors of the Fed-I-AMP model remain at a relatively low level. Taking MAE in Scenario II as an example, the prediction error of Fed-I-AMP is merely 0.0999, which represents a slight improvement over Solo (0.1213) and is significantly lower than FedAVG (0.2298) and FedProx (0.2220) under the same conditions. Moreover, it outperforms FedAMP (0.1497). This outcome validates the advantages of the proposed method in learning from non-IID data. This is attributed to the integration of EWC algorithm into the model local training process, which encourages each local model to converge to a similar parameter space, thereby effectively overcoming the model performance degradation due to parameter differences during aggregation. Furthermore, when dealing with larger datasets, the prediction accuracy and convergence performance of the proposed method remain significantly higher than those of traditional methods such as FedAvg and FedProx, and it also shows

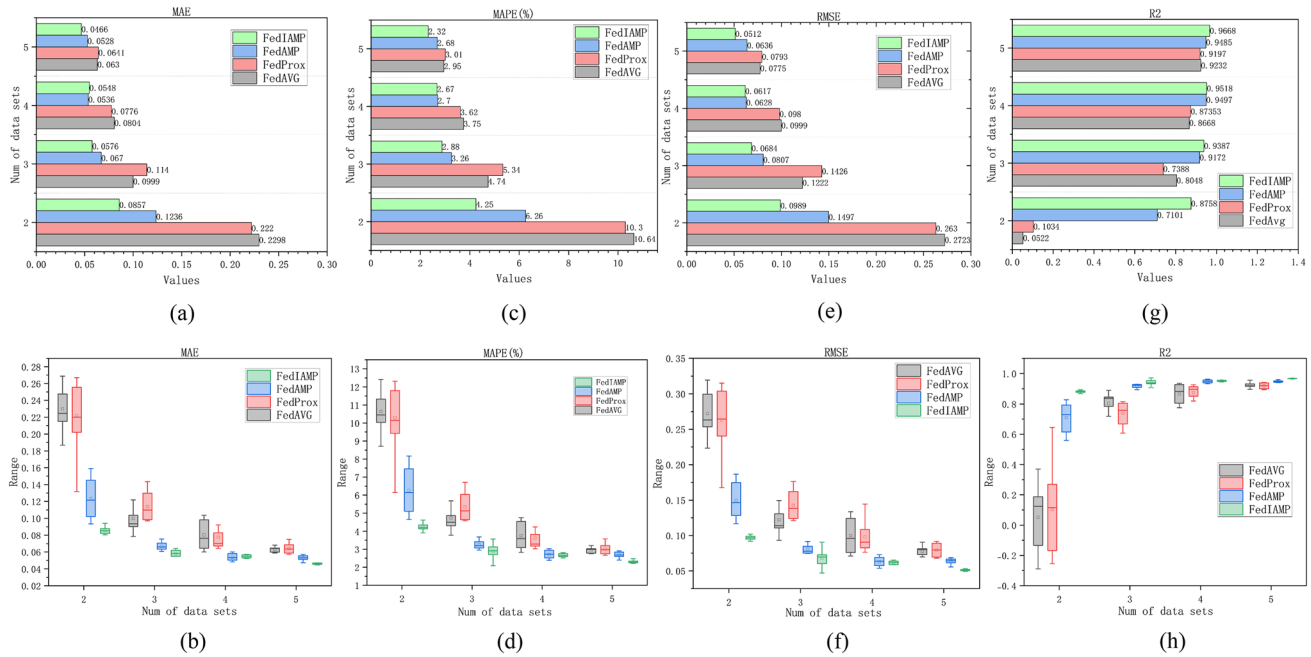

Fig. 9 Performance comparison at different data sizes

advantages over FedAMP, validating the effectiveness of the improvement strategies presented.

In summary, the increase in data size has a beneficial impact on the convergence and accuracy of FL models. Notably, the proposed method outperforms the other three traditional FL methods in terms of predictive performance and accuracy across various dataset sizes, particularly in smaller datasets. Our method achieves the highest predictive performance in Scenario Five, with MAE, Mean MAPE, RMSE and R^2 values of 0.0466, 2.32%, 0.0512, and 0.9668, respectively, demonstrating the superiority of this paper's method.

5.4 Optimization of Hobbing Process Parameters

To obtain the optimal process parameters for low-carbon hobbing in the target enterprise, the MOCOA was employed to solve the above multi-objective optimization model. The code was implemented in Python, where the population size of coatis was set to 100, the maximum number of iterations was set to 100, and the grid number was configured as 10. It is noteworthy that although the input dimension of the carbon emissions prediction model is 8, only four controllable variables, namely the spindle speeds and spindle feeds in the rough and finish hobbing stages, are practically adjustable process parameters. Therefore, the characteristic input dimension of the population is 4. The process parameters obtained by solving with MOCOA are shown in Table 7.

Furthermore, to validate the practicality and effectiveness of the optimized results, a comparison was conducted between the optimal process parameters derived under several scenarios and the empirical schemes. Scenario 1 represents the empirical scheme, Scenario 2 involves a single-objective optimization of the model that considers solely the hobbing carbon emissions, Scenario 3 focuses on a single-objective optimization targeting solely the CEI, and Scenario 4 encompasses a multi-objective optimization that comprehensively considers both carbom emission (C) and CEI.

The optimization results under different optimization strategies reveal a complex relationship between C and the CEI in hobbing. Specifically, optimizing solely for C can lead to a reduction of up to 11.63% in C, accompanied by a decrease of 12.59% in CEI. This is primarily attributed to the fact that lower spindle speeds and higher feed rates effectively reduce the machining power and time of the hobbing process, thereby decreasing carbon emissions. Simultaneously, such process parameter settings enhance production efficiency and indirectly lower production costs, resulting in a certain degree of reduction in CEI. Therefore, C and CEI are not entirely opposing indicators; rather, they exhibit a positive correlation within a certain range.

In contrast to the empirical scheme, optimizing the CEI solely resulted in a significant reduction of 25.11% in CEI, but a marginal increase of 0.5% in C. Compared to Scenario 2, the process parameters adopted in this case were relatively conservative, with increased Spindle speeds and decreased feed rates, which facilitated the production of high-precision

Table 7 Optimization results table

No	Optimization objective	n_1	f_1	n_2	f_2	C	CEI
1	EC	320	4	390	6	2.2541	0.9270
2	Min C	280	4.5	360	8	1.9920	0.8103
3	Min CEI	360	4.5	401	7.1	2.2654	0.6942
4	Min (C, CEI)	360	4.5	381	8	2.0968	0.7180

gears and reduced hob wear, thereby lowering costs. Consequently, the decrease in CEI was pronounced. However, this resulted in an increase in C, indicating a certain degree of conflict between the two indicators within a specific range.

The above analysis highlights the complex relationship between C and CEI, where single-objective optimization schemes struggle to achieve a satisfactory balance between the two. Further examination of multi-objective optimization schemes shows that, compared to the empirical approach, the multi-objective optimization results in a 6.08% reduction in C and a 22.55% decrease in CEI. When compared to the scenario where only C are optimized, although there is a marginal increase in C (5.26%), there is a more pronounced improvement in CEI (11.39%). Additionally, in comparison to the scenario where only CEI is optimized, despite a slight increase in CEI (3.42%), a significant decline in carbon emissions (7.44%) is discernible.

In summary, when enterprises pursue conflicting indicators such as emission reduction and comprehensive benefits, the multi-objective optimization method proposed in this paper can assist in achieving a balance between these objectives, thereby avoiding the limitations of single-objective optimization. It is crucial to note that the weight parameters for CEI in this study were determined using the objective weighting method of entropy weight approach. However, in practical applications, enterprises can adjust the weights based on their own strategic goals to better align with actual demands and flexibly modify process parameters according to the optimization outcomes.

6 Conclusions

This paper proposes a method for collaboratively training a hobbing carbon emissions prediction model using multi-source data based on federated learning, and further optimizes the hobbing process parameters. This method leverages data from similar enterprises to collaboratively train the model while safeguarding the data privacy of collaborating enterprises through model sharing. Furthermore, upon establishing the prediction model, this paper reduces the carbon emissions and comprehensive efficiency index of hobbing processes by optimizing the process parameters. Experimental results show that this method can effectively mitigate the challenges posed by the scarcity of manufacturing data in a single enterprise for model training and optimization, thereby providing scientific guidance for practitioners to optimize hobbing process parameters based on data-driven approaches for energy conservation and emission reduction.

A comparative analysis across multiple scenarios shows that a model trained using multi-source data exhibited significantly enhanced prediction accuracy compared to models trained on a single dataset. Furthermore, the proposed Fed-I-AMP method shows superior prediction accuracy and robustness in non-IID hobbing datasets, outperforming traditional FL approaches. Further optimization results under diverse conditions indicate that achieving low-carbon hobbing operations often necessitates higher spindle speeds and lower feed rates. However, radical adjustments can result in an increase in CEI, thereby undermining the overall corporate benefits. By utilizing a multi-objective optimization algorithm, a balance can be achieved between carbon emissions and CEI. By setting optimized hobbing process parameters, a 6.08% reduction in carbon emissions and a 22.55% decrease in CEI were achieved compared to traditional empirical methods.

Although this study contributes to the realization of low-carbon hobbing processes, it is limited by experimental conditions and does considerate factors such as plant temperature and machine tool vibration that may impact hobbing carbon emissions. In future research, the intention is to continue to collect production data from various types of hobbing processes on different machine tools across different enterprises, aiming to continually expand the data volume and quality of the multi-source carbon emissions dataset, with the goal of developing a high-precision prediction model for hobbing processes in more complex scenarios.

Acknowledgements This work was supported by the National Natural Science Foundation of China (No. 52005062) , the Natural Science Foundation of Chongqing Municipality (CSTB2023NSCQMSX0390) and Foundation of State Key Laboratory of Public Big Data (PBD2024-0512).

Author Contributions Qian Yi: Conceptualization, Methodology, Resources, Writing – review and editing, Funding acquisition. Yan Xu: Software, Formal analysis, Investigation, Validation, Writing – original draft. Congbo Li: Supervision, Project administration. Chuanjiang Li: Data Curation. Huajun Cao: Supervision, Project administration.

Funding The National Natural Science Foundation of China,No. 52005062, Qian Yi,the Natural Science Foundation of Chongqing

Municipality, CSTB2023NSCQMSX0390, Qian Yi, Foundation of State Key Laboratory of Public Big Data, PBD2024-0512, Qian Yi.

Data Availability The complete code is uploaded to the GitHub repository and found at the following link https://github.com/xy2000ha/FedIAMP.

Declarations

Conflict of interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Input: neural network parameters [w_1, w_2, …, w_k, w_t], Gear hobbing carbon emission datasets [D_1, D_2, …, D_k], maximum number of iterations: Iter, Number of client servers: K, Maximum number of communication rounds: Cr

Output: neural network parameters [w_t, w_2, …, w_k]

```
Randomly initialize [w_1, w_2, …, w_k] on clients.
while r ≤ Cr do:
  For epoch =1 to Iter do:
    loss = mse_loss (w_1, D_1)
    Update target model parameters w_1 by backward propagation algorithm
  End for
  For i = 2, …, K do:
    w_i = w_1
    For epoch =1 to Iter do:
      Calculate ewc_loss on D_i by Eq. (11)
      Total_loss = mse_loss (w_i, D_i) + ewc_loss
      Update collaborator model parameters w_i by backward propagation algorithm
    End for
  End for
  For k = 1, 2, …, K do:
    Cloud servers collect [w_1, w_2, …, w_k] to calucate [u_1, u, …, u_k] by Eq. (17)
    Each client to compute w1 by Eq. (18)
  End for
  r = r + 1
End while
```

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